

- 8 (a) (i) number = $(5.1 \times 10^{-6} \times 6.02 \times 10^{23}) / 241$ C1
 = 1.27×10^{16} A1 [2]
- (ii) $A = \lambda N$ C1
 $5.9 \times 10^5 = \lambda \times 1.27 \times 10^{16}$
 $\lambda = 4.65 \times 10^{-11} \text{ s}^{-1}$ A1 [2]
- (iii) $4.65 \times 10^{-11} \times t_{1/2} = \ln 2$ C1
 $t_{1/2} = 1.49 \times 10^{10} \text{ s}$
 = 470 years A1 [2]
- (b) sample / activity would decay appreciably whilst measurements are being made B1 [1]

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Section B

0 (c) (i) fraction of the output (signal) is added to the input (signal)

M1